

StockVision - Problem Statement

Problem Statement (V2) | Scilab GUIVerse Hackathon

1. The problem

I saw a gap between introductory Scilab examples and an interactive application that lets a learner explore a realistic predictive workflow. Stock-price prediction is often presented as a black box: data goes in and a number comes out, while the user has little chance to see how modelling choices, training history, validation design and trading assumptions change the answer - or whether the model beats a trivial guess at all.

Stock-market modelling naturally combines data visualization, forecasting, comparison, validation and decision-oriented analysis in one GUI workflow, which fits the hackathon's focus on original, interactive Scilab applications.

2. The gap I wanted to fill

I wanted a lightweight Scilab application in which a user can answer questions such as:

- How does a feature-based Linear Regression compare with an autoregressive model and with Holt exponential smoothing on the **same data and the same test window**?
- Does any of them beat a **naive "tomorrow equals today" baseline**? If not, the application should say so.
- How do results change when the train/test split, the AR lookback or the number of validation folds changes?
- Do the models hold up under **walk-forward validation** rather than a single split?
- What happens to a model-guided signal once a one-day execution lag, transaction costs and slippage are applied, and how does it compare with **buy and hold** over the same period?
- Is the data itself trustworthy (missing values, duplicates, ordering, impossible bars)?

Normally this requires writing code, running separate experiments and comparing outputs by hand. I wanted a single interactive workflow.

3. What I built

StockVision is a Scilab GUI organised as **one integrated dashboard**: controls on the left; the price/prediction chart, model-comparison chart and equity curve embedded in the same window; and panels for data quality, model results, metric comparison with ranking, walk-forward folds, backtest summary and backtest assumptions. Nothing opens in a separate window.

The models are Linear Regression (12 engineered features), an AR time-series model, Holt exponential smoothing and a naive baseline. All are evaluated with RMSE, MAE, MAPE, R^2 and directional accuracy; the ranking is computed from the actual results. Model Info explains each model's purpose, assumptions and limitations in plain language.

4. Intended users and use cases

Students and learners of regression, time-series modelling, smoothing methods, data visualization and basic financial analysis. A user can compare modelling approaches, see why a baseline matters, test how parameter choices affect results, understand evaluation metrics, and see how a simple historical strategy behaves under explicit assumptions.

5. Why this approach

I chose depth of evaluation over breadth of features. Every control changes a real modelling, validation or backtesting input; every displayed number is computed from the data; and the application deliberately reports unflattering outcomes. On the bundled synthetic datasets, for example, none of the three models achieves a lower test RMSE than the naive baseline - which is an honest and educational result rather than a defect to hide.

6. Scope, limitations and responsible use

- The bundled datasets are **synthetic demonstration data**; there is no live market connection.
- StockVision is an educational and analytical tool. It is **not investment advice** and does not claim to predict markets or generate profit. Past or simulated performance does not indicate future results.
- Backtests cover short windows, are long/cash only, and ignore bid/ask spread, partial fills, intraday movement and market impact; annualized figures from short windows are extrapolations.
- Models are simple linear/smoothing methods with fixed or user-chosen parameters; there is no deep learning and no external AI service.

- Metrics are noisy on small samples and no statistical significance testing is performed.

7. Expected outcome

My objective is not to claim that a model can reliably predict financial markets. It is to make the modelling process understandable and testable: compare approaches against a baseline, question parameter choices, validate forward in time, and see how a simple strategy behaves under explicit assumptions.

In one sentence: I built StockVision to turn stock forecasting from a single black-box number into an interactive, honest Scilab workflow for comparison, validation, visualization and practical evaluation.